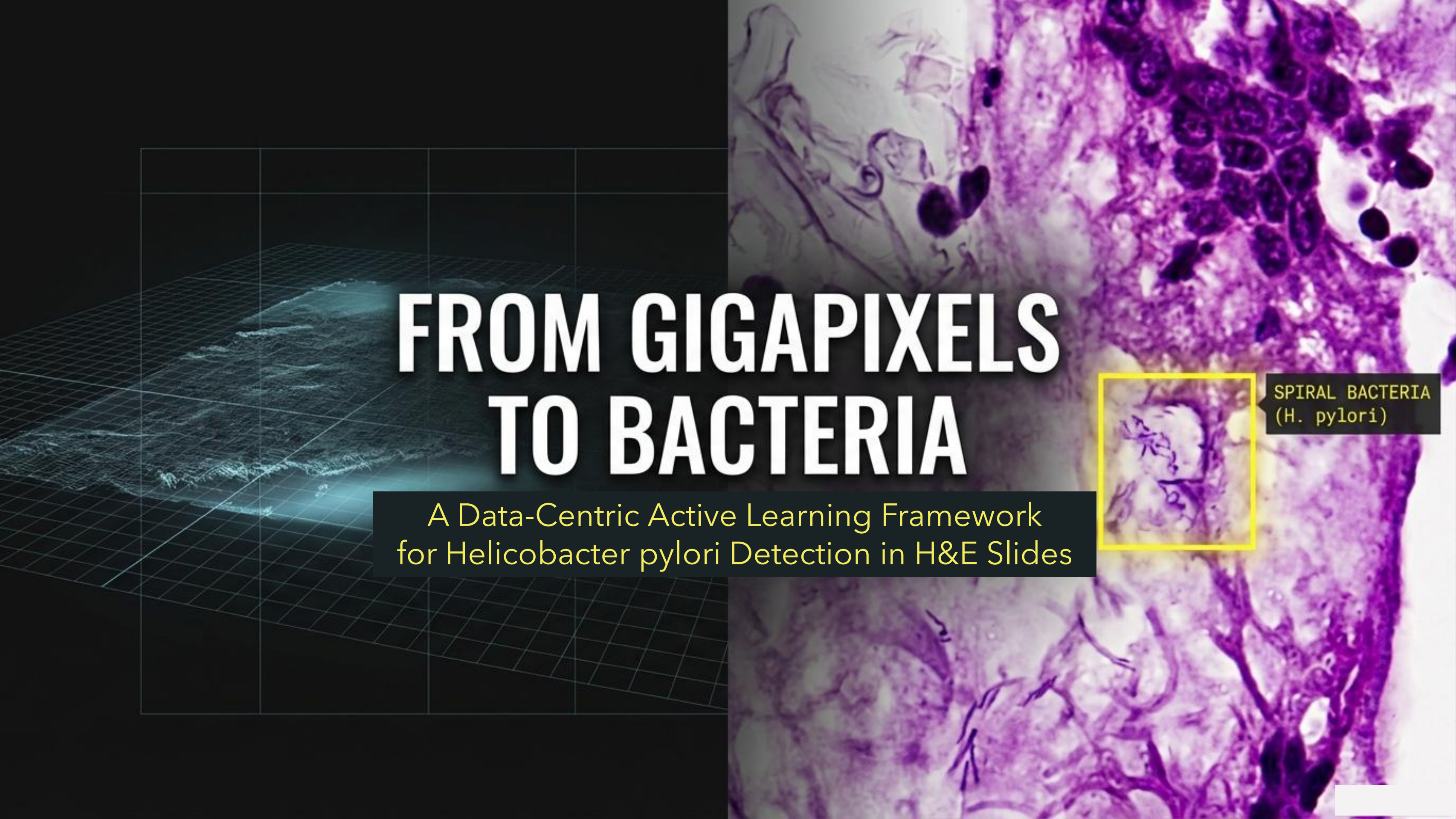




FROM GIGAPIXELS TO BACTERIA

A Data-Centric Active Learning Framework
for *Helicobacter pylori* Detection in H&E Slides



SPIRAL BACTERIA
(*H. pylori*)

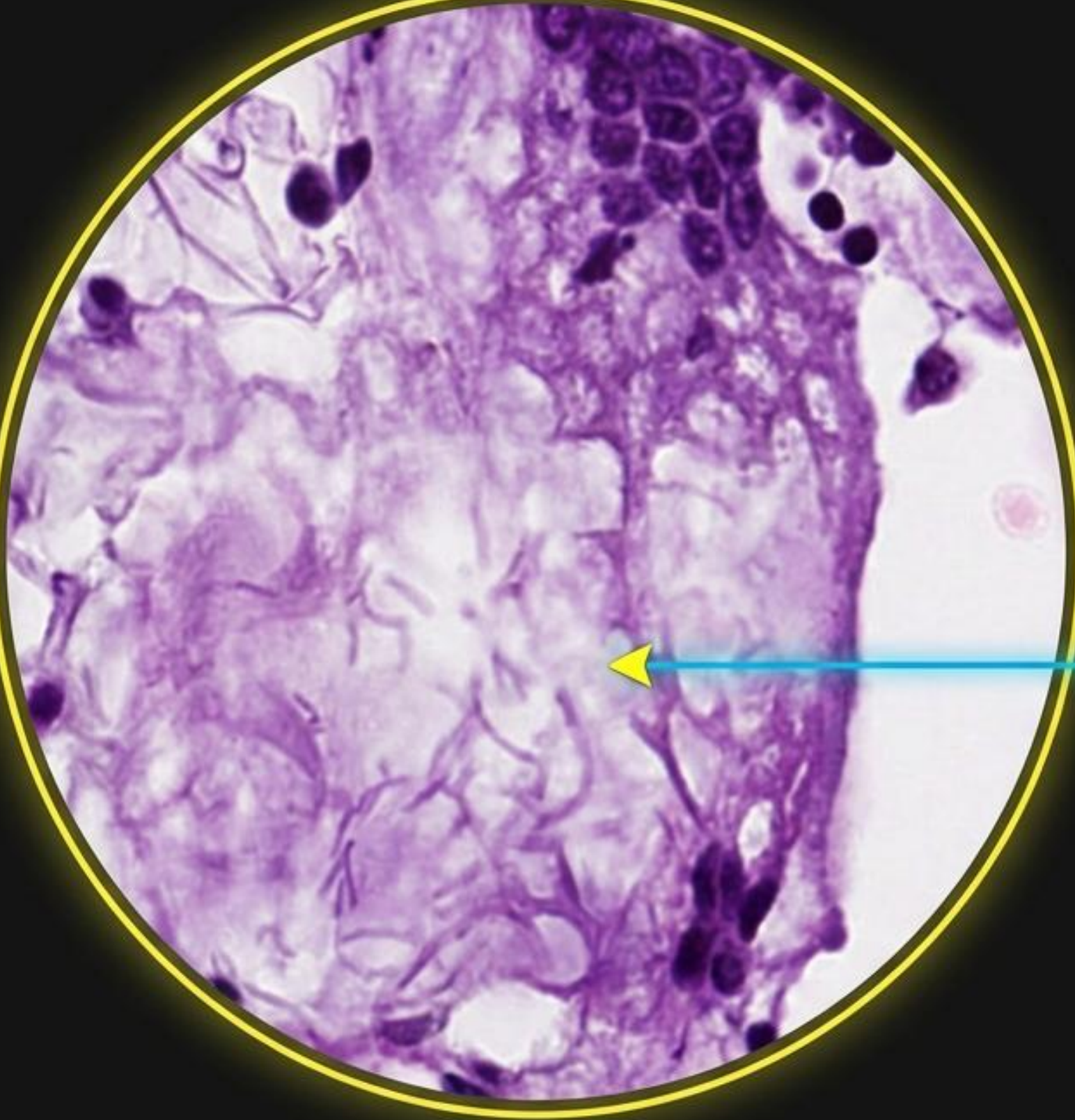
THE SUSPECT: HELICOBACTER PYLORI

TARGET: Class I Carcinogen

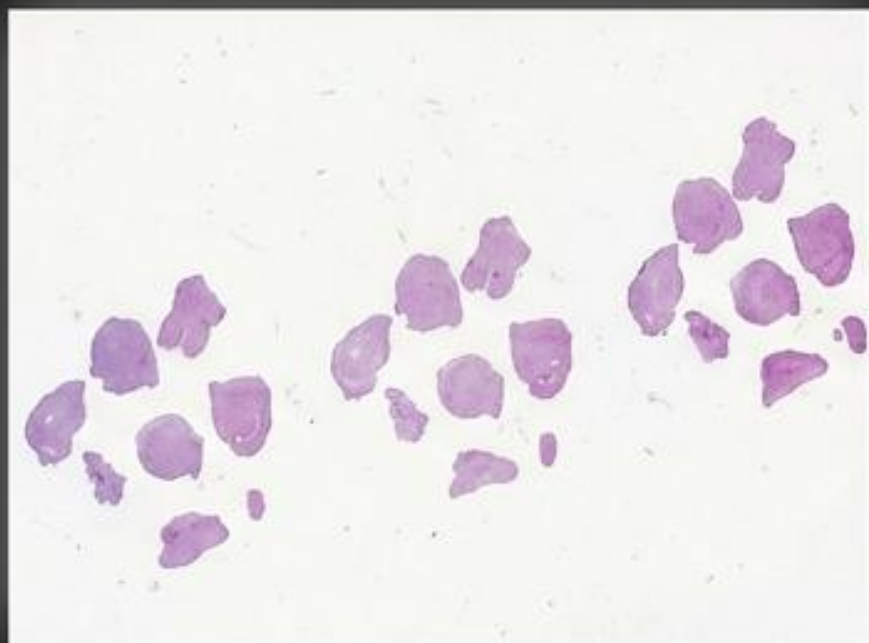
SIZE: 2–5 μm (Microscopie)

LOCATION: Mucus layer &
gastric glands

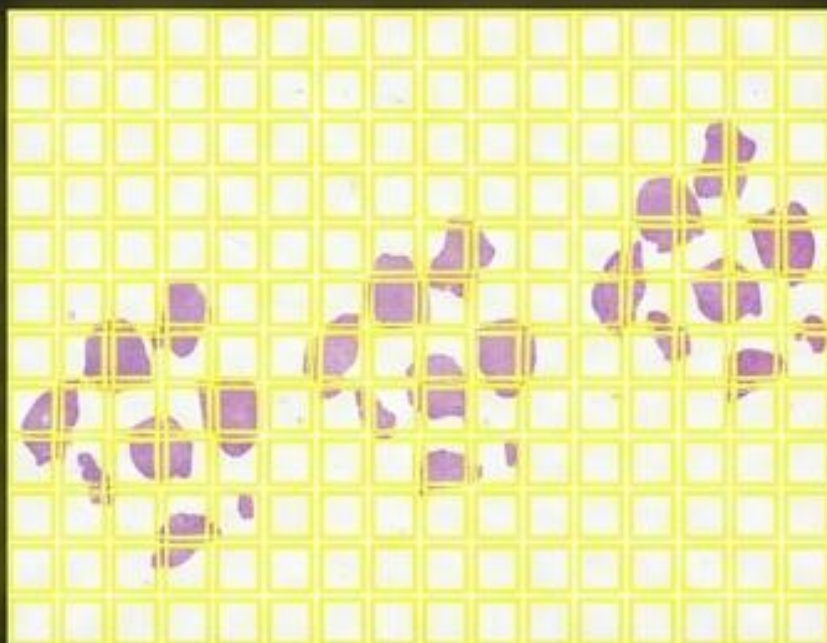
CONTEXT: Veterinary Biopsies
(Chronic Gastritis)



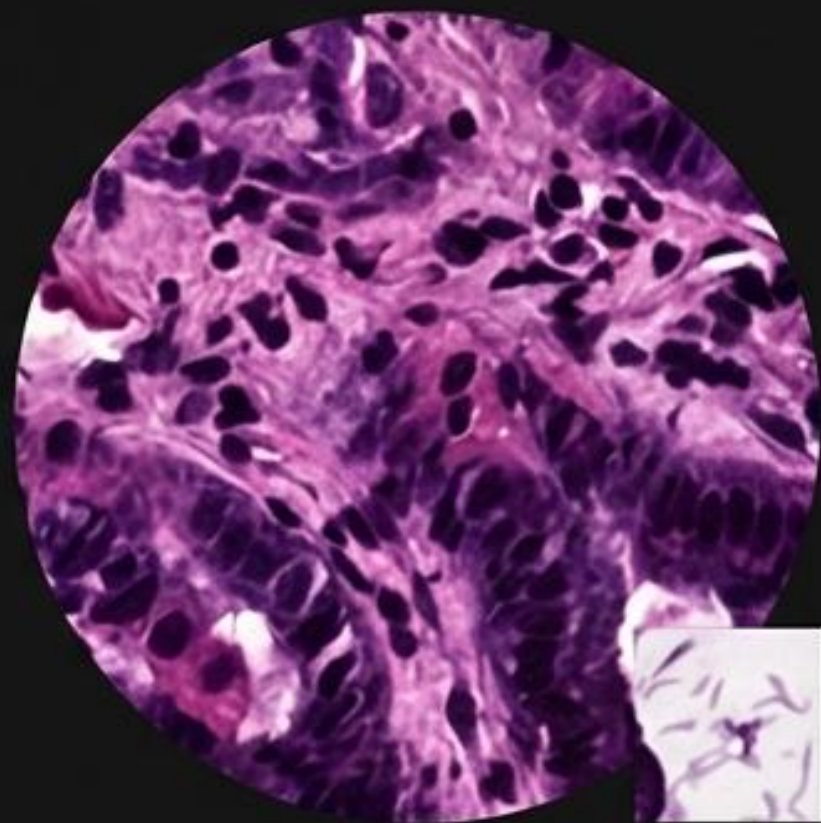
DATA SCALE: THE NEEDLE IN THE HAYSTACK



1 GIGAPIXEL WSI



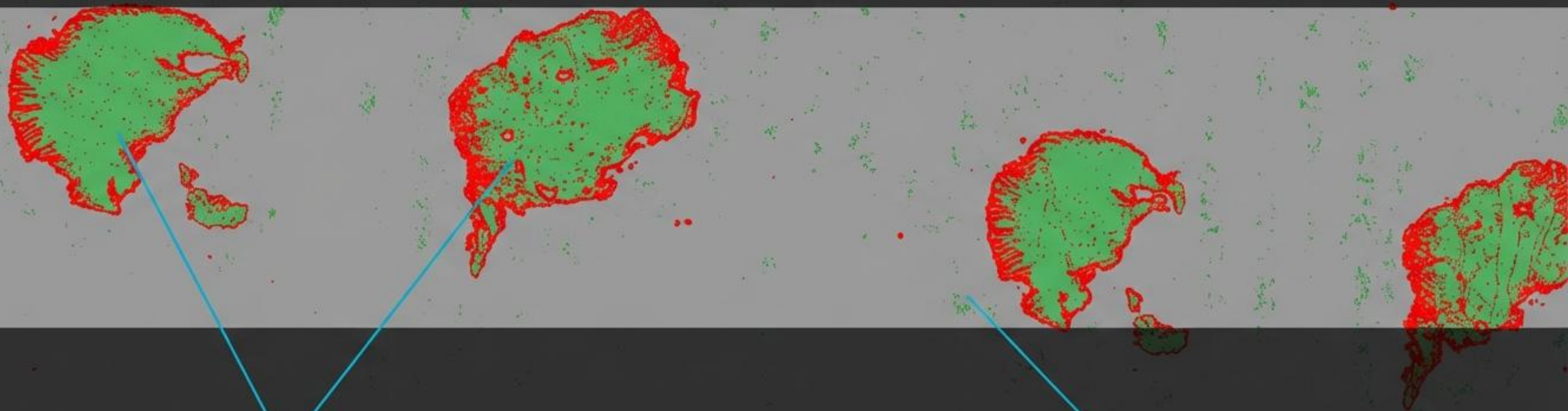
~100,000 PATCHES



512 x 512 PIXELS

MANUAL SEARCH: ~30 MINS PER SLIDE

NARROWING THE SEARCH



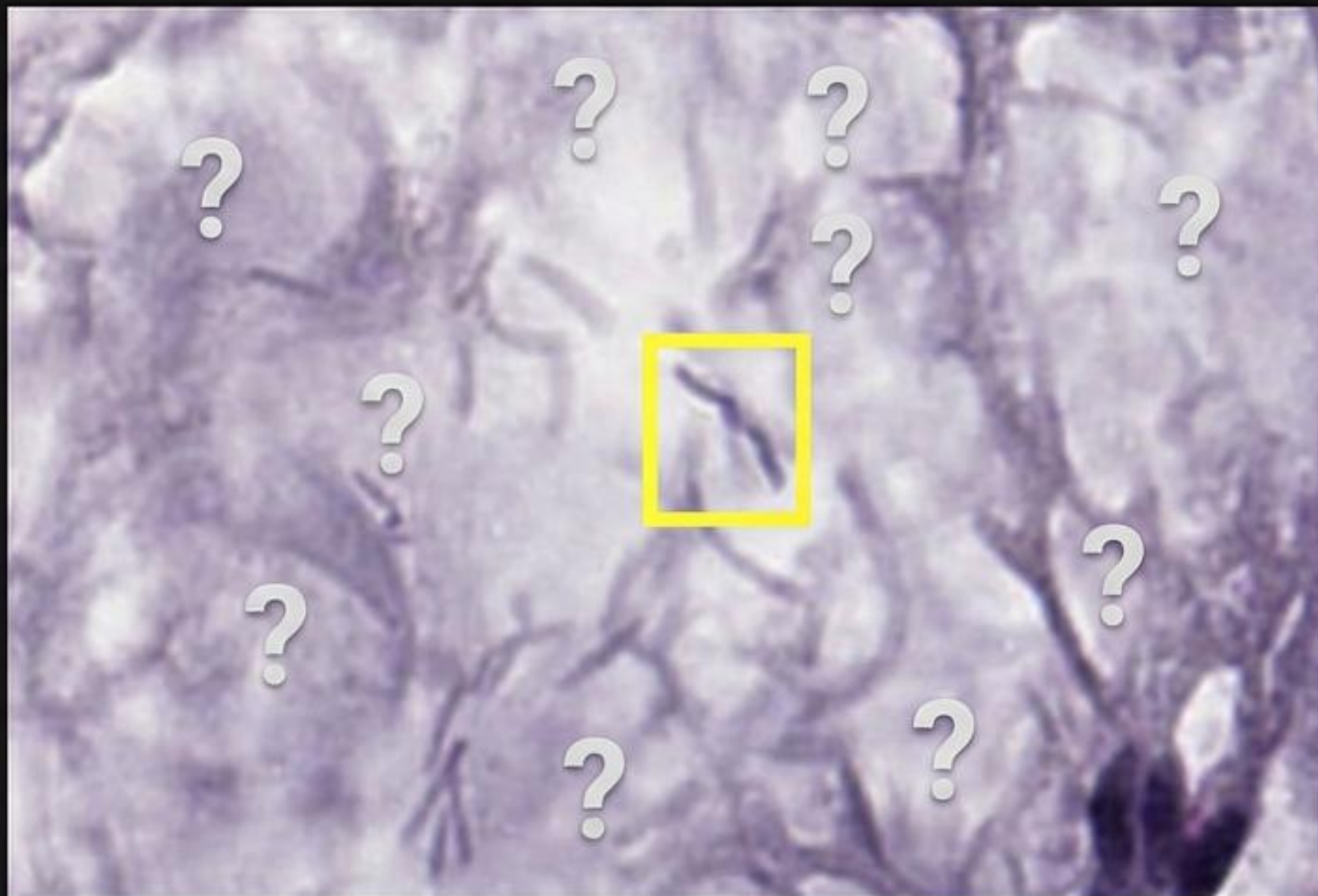
HOTSPOTS
(Glandular Epithelium)

IGNORED
(Stroma & Background)

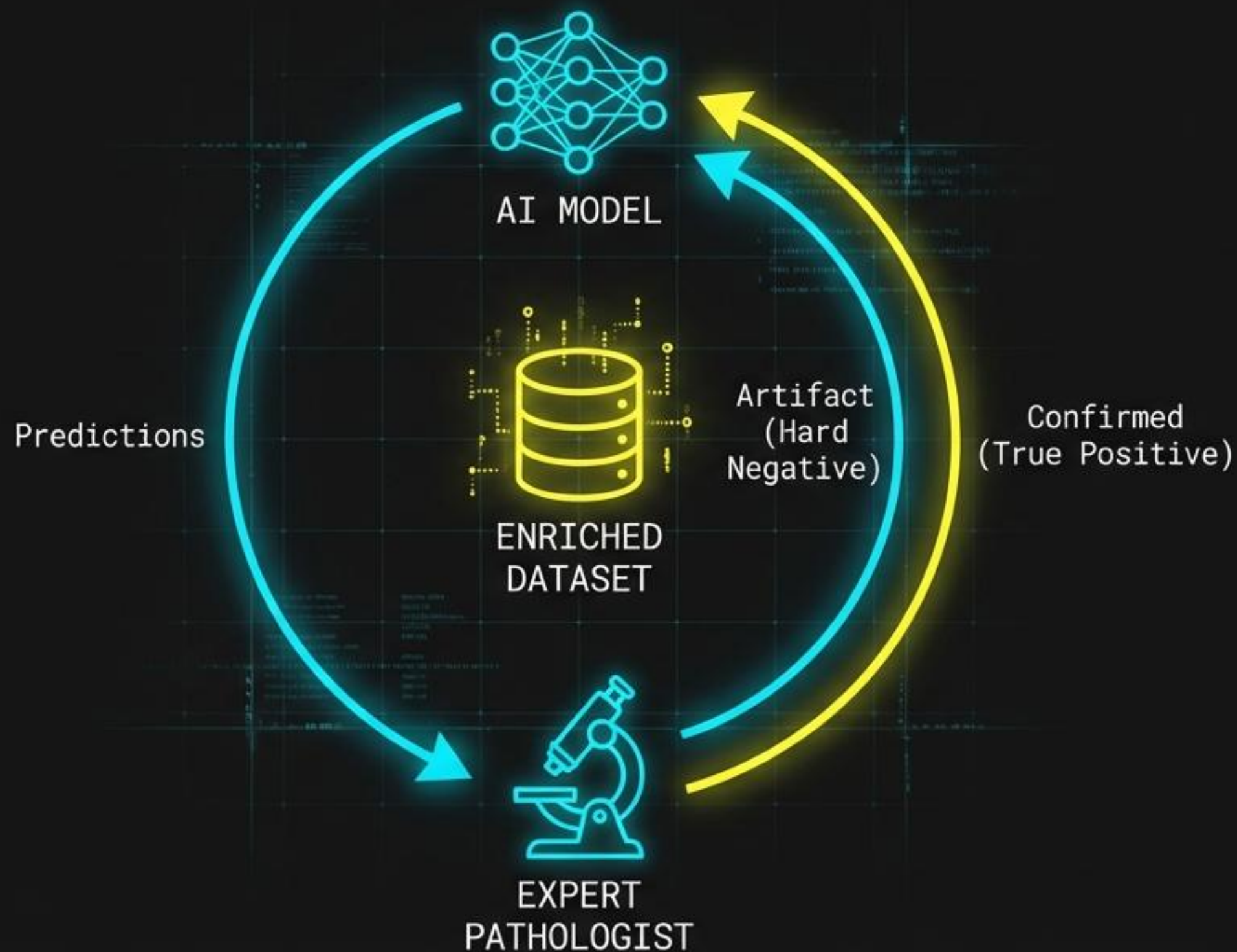
QuPath Pre-processing reduces search space by 60%

THE SILENT MAJORITY

The Data Problem: Incomplete Annotations.
To the Model, "Unlabeled" = "Negative".



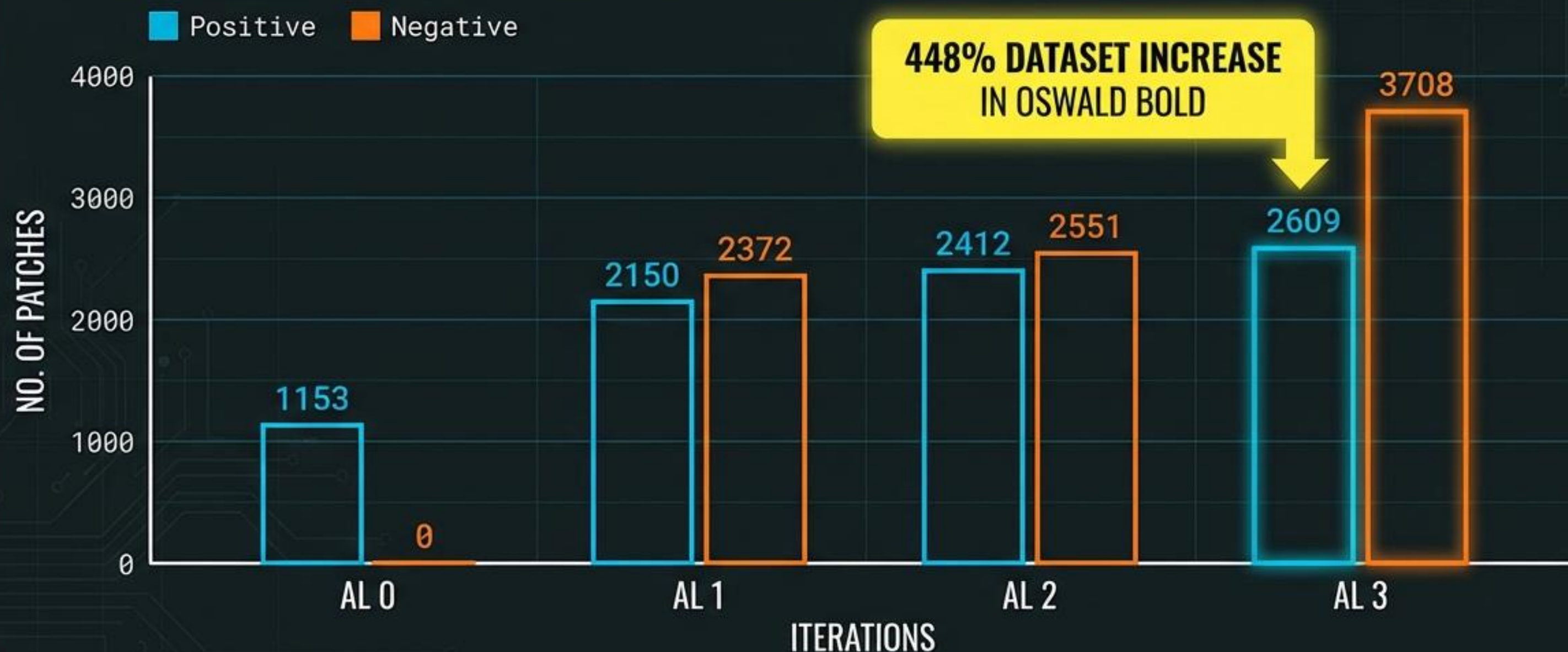
THE INTERROGATION



ITERATION 3: THE BREAKTHROUGH

Teaching the model exactly what is NOT bacteria.

DATASET GROWTH ACROSS ITERATIONS

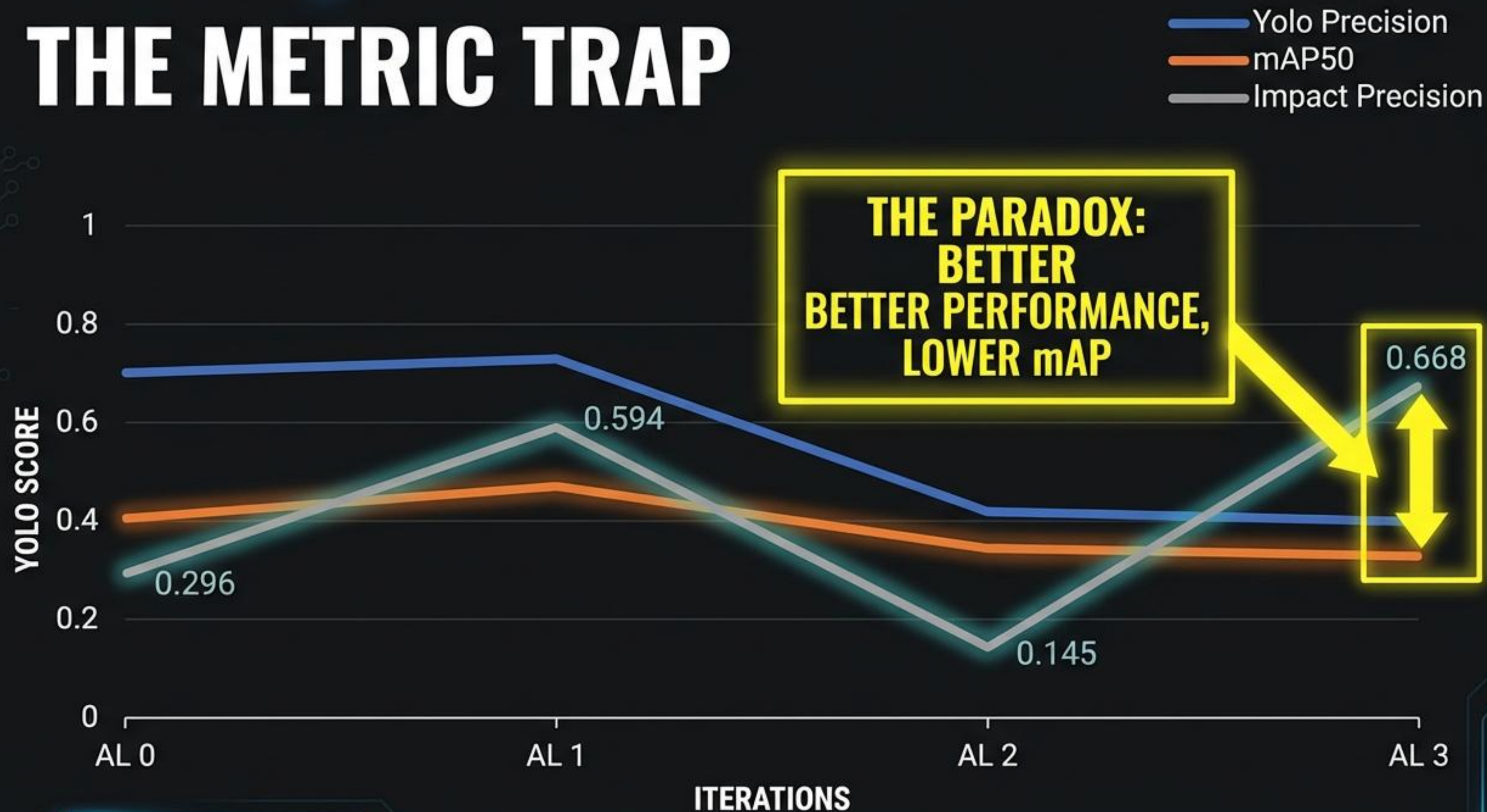


A NEW RULER

$$\text{Impact Precision} = \frac{\text{Expert Confirmed Bacteria}}{\text{Total AI Predictions}}$$

80% Precision = 4 out of 5 AI flags are real infections.

THE METRIC TRAP



THE SCOUT

YOLOv8

- Role: Active Learning
- Strength: High Recall
- Task: Find all candidates.

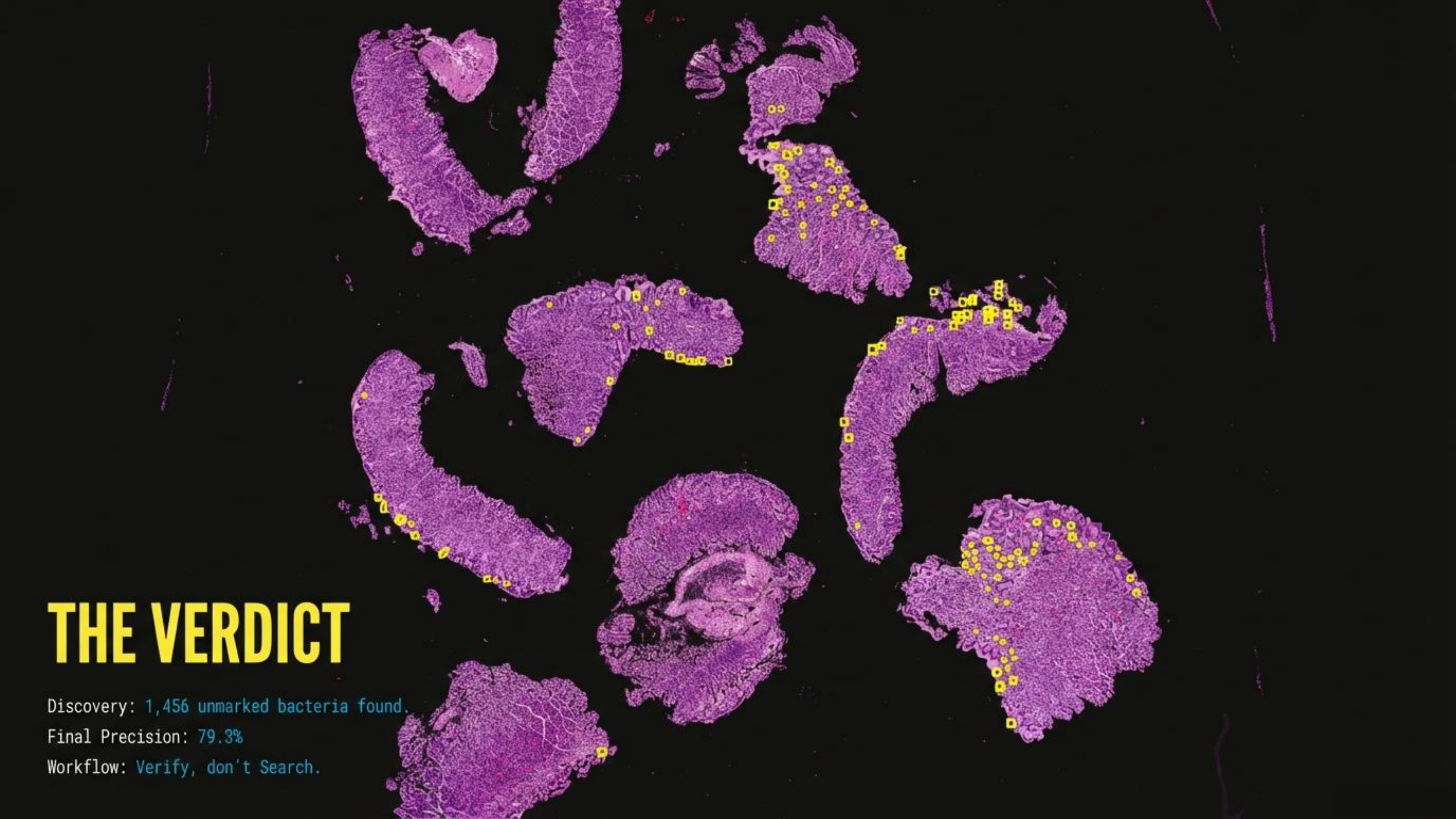


THE SPECIALIST

Faster R-CNN

- Role: Final Deployment
- Strength: Two-Stage Precision
- Task: Verify morphology.





THE VERDICT

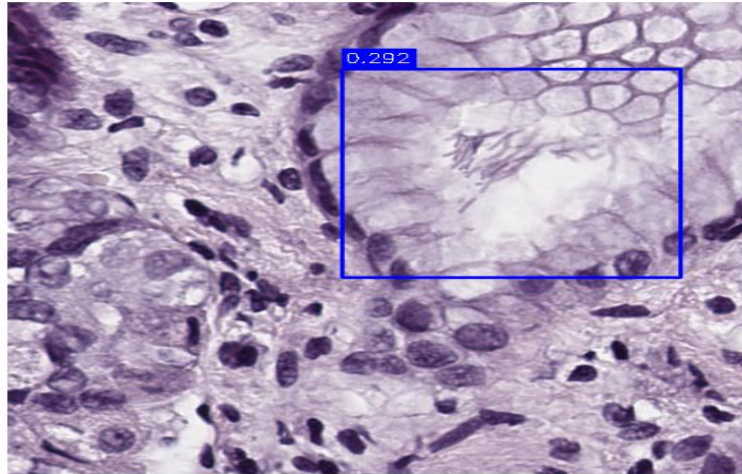
Discovery: 1,456 unmarked bacteria found.

Final Precision: 79.3%

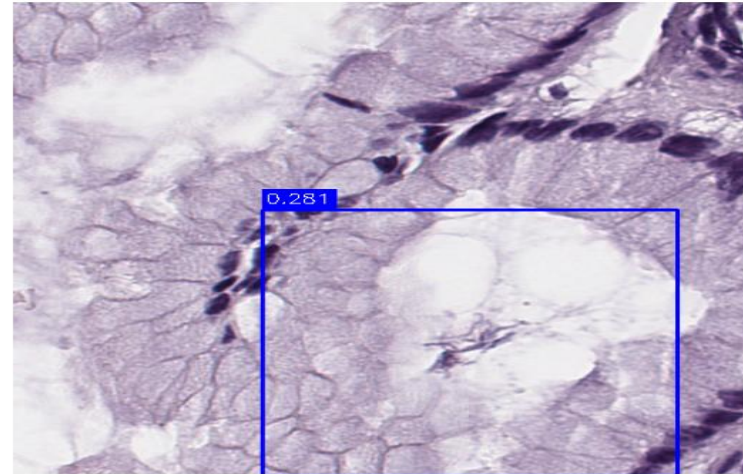
Workflow: Verify, don't Search.

DETECTIONS SHOWN

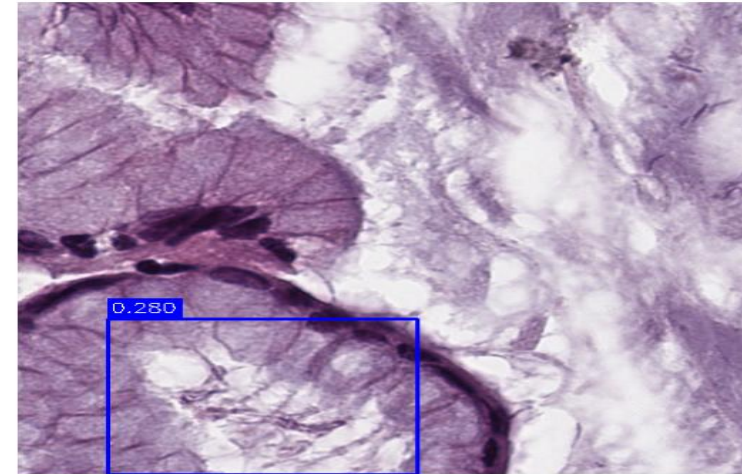
522934_x129720_y29440
1 bacteria (max: 0.292)



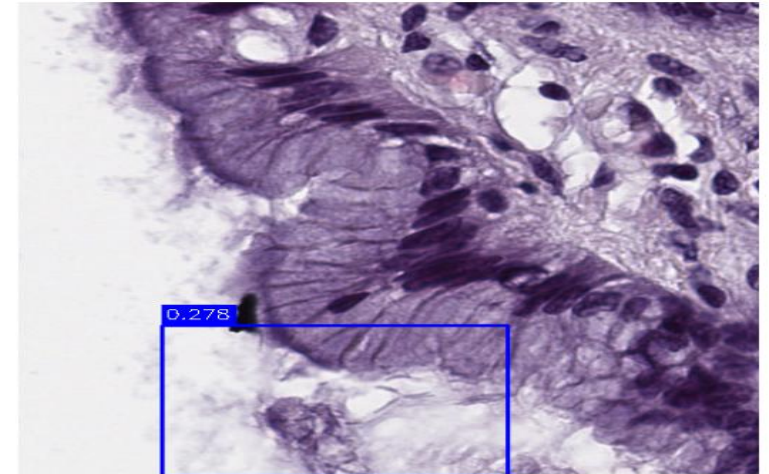
522934_x129260_y44620
1 bacteria (max: 0.281)



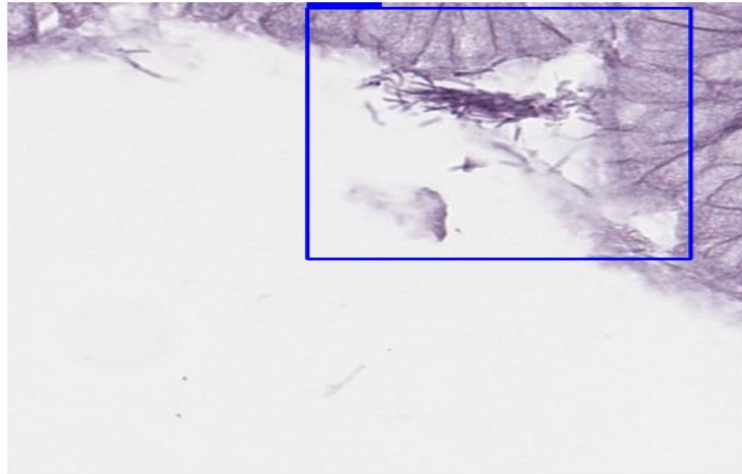
522934_x81880_y42320
1 bacteria (max: 0.280)



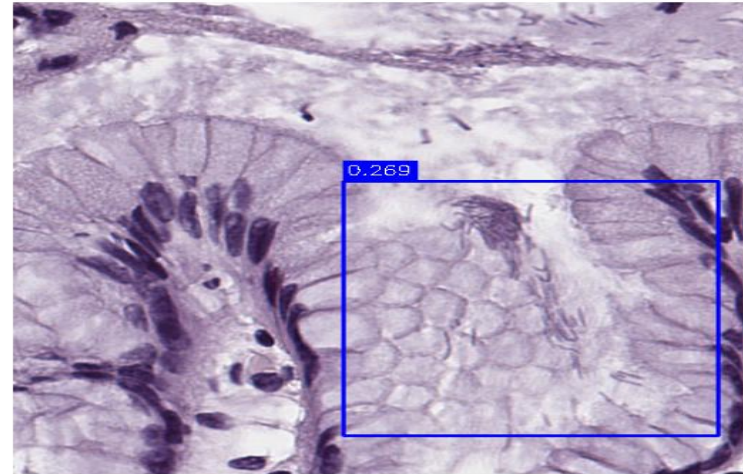
522934_x50140_y26220
1 bacteria (max: 0.278)



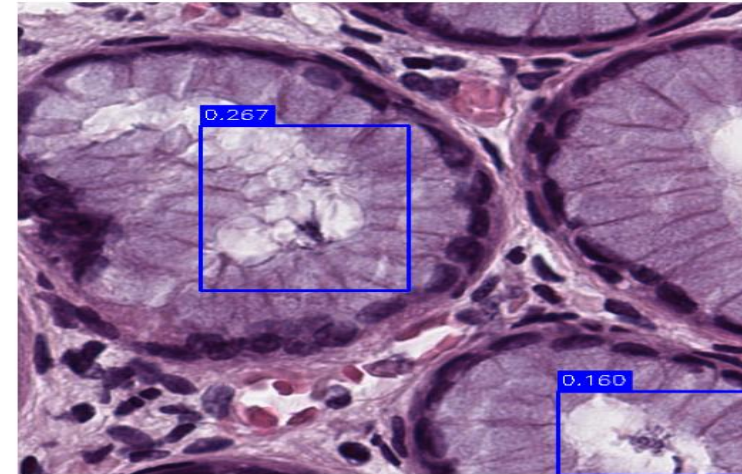
522934_x114540_y30820
1 bacteria (max: 0.272)



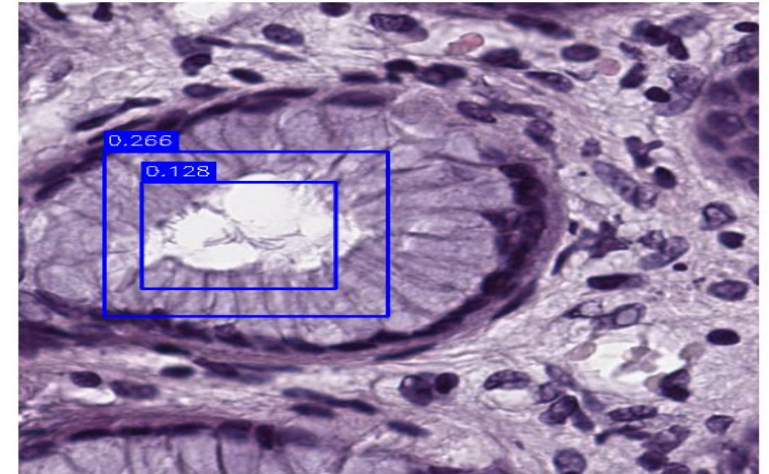
522934_x128800_y28980
1 bacteria (max: 0.269)



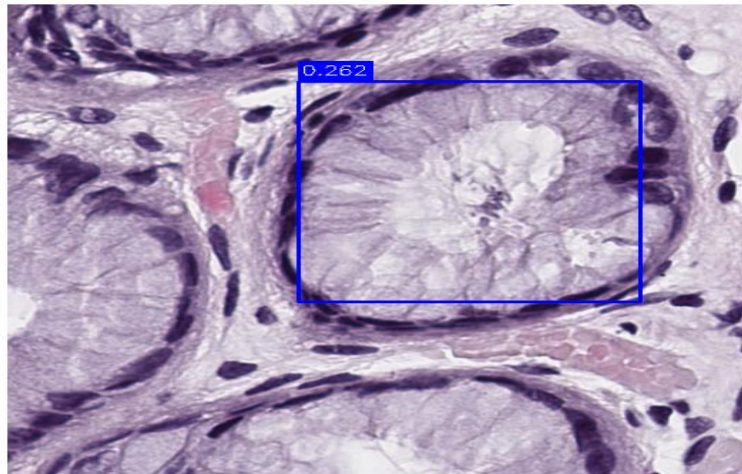
522934_x76360_y39100
2 bacteria (max: 0.267)



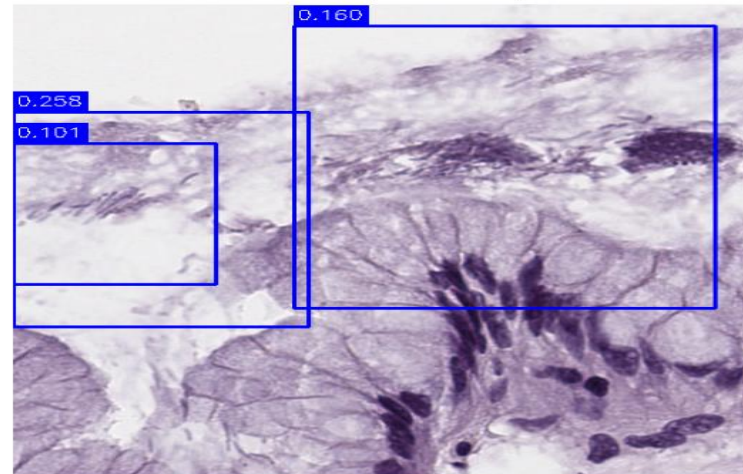
522934_x72220_y17020
2 bacteria (max: 0.266)



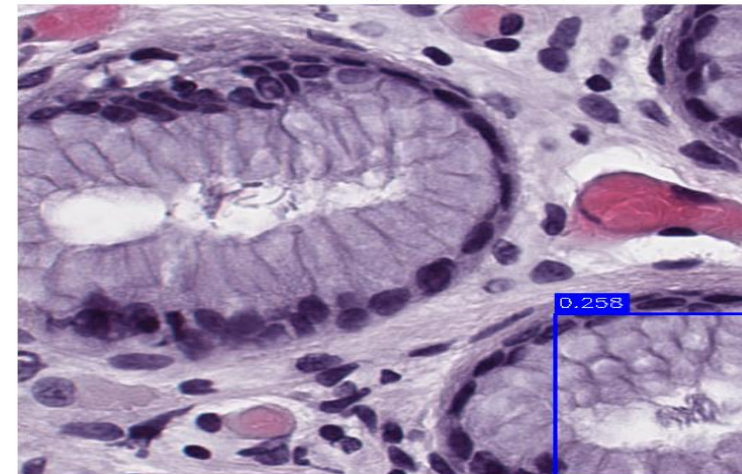
522934_x129720_y44620
1 bacteria (max: 0.262)



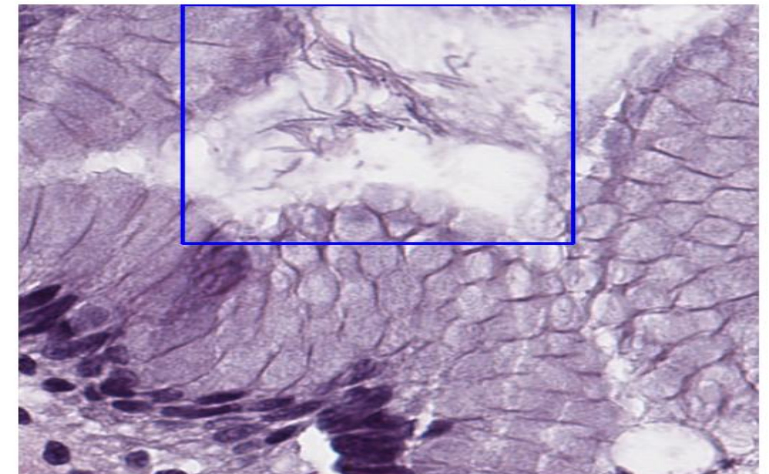
522934_x127880_y28980
3 bacteria (max: 0.258)



522934_x169280_y23000
1 bacteria (max: 0.258)



522934_x176640_y38640
1 bacteria (max: 0.255)





IT'S NOT THE CODE. IT'S THE DATA.

Hard Negatives are as valuable as Positives.

THANK YOU

Questions?

ACKNOWLEDGEMENTS

Supervisor: Prof. Dr. Magda Gregorová | THWS, Würzburg

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